

InstCycle	$\frac{\neg \text{acyclic}(R_1, \dots, R_k) \in \Delta^* \quad ((R_1, \dots, R_k), _, _) \notin C}{C := C \cup \{((R_1, \dots, R_k), s, 1)\}$ $\Delta := \Delta \cup \{\text{len}(s) > 1, \text{minimal}((R_1, \dots, R_k), s)\}$
SplitCycleLen	$\frac{((R_1, \dots, R_k), s, \text{cnt}) \in C}{\Delta := \Delta \cup \{\text{cnt} < \text{len}(s)\} \quad \parallel \quad \Delta := \Delta \cup \{\text{cnt} \approx \text{len}(s)\}}$
UnrollCycle	$\frac{((R_1, \dots, R_k), s, \text{cnt}) \in C \quad \text{cnt} < \text{len}(s) \in \Delta^*}{C' \equiv C \setminus \{((R_1, \dots, R_k), s, \text{cnt})\} \cup \{((R_1, \dots, R_k), s, \text{cnt}+1)\}}$ $\begin{aligned} C &:= C' & \Delta &:= \Delta \cup \{(s[\text{cnt}-1], s[\text{cnt}]) \in R_1^+\} \\ & & & \parallel \dots \\ & \parallel C &:= C' & \Delta := \Delta \cup \{(s[\text{cnt}-1], s[\text{cnt}]) \in R_k^+\} \end{aligned}$
ContrMinimal	$\frac{((R_1, \dots, R_k), s, \text{cnt}) \in C \quad \text{minimal}((R_1, \dots, R_k), s) \quad b \in [1, k] \quad 1 \leq q < r-1 \leq \text{len}(s) \quad (s[q], s[r]) \in R_b^+}{\text{unsat}}$

Example 1: SAT instance

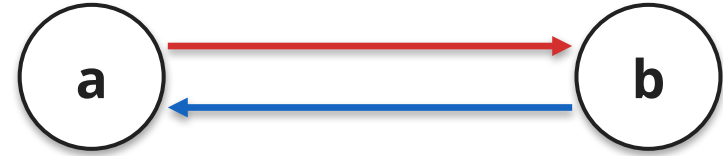
R1 = {(a, b)}

R2 = {(b, a)}

R3 = {(c, d)}

$\neg \text{acyclic}(\text{R1}, \text{R2}, \text{R3})$

i.e., $R1 \cup R2 \cup R3$ contains a cycle



Example 1: derivation

Step 1 — InstCycle

Premise(s): $\neg \text{acyclic}(R1, R2, R3)$

$\Rightarrow C = \{((R1, R2, R3), s, 1)\}$

$\Delta = \{\text{len}(s) > 1, \text{minimal}(s)\}$

$$\Delta_0 = \{\text{len}(s) > 1, \text{minimal}(s)\}$$

Step 2 — SplitCycleLen

Premise(s): $((R1, R2, R3), s, 1) \in C$

$\Rightarrow \Delta = \Delta \cup \{1 < \text{len}(s)\} \quad || \quad \Delta = \Delta \cup \{1 = \text{len}(s)\}$

(we only explore case 1)

$$\Delta_0' = \{\text{len}(s) > 1, \text{minimal}(s)\}$$

Step 3 — UnrollCycle

Premise(s): $((R1, R2, R3), s, 1) \in C$ and $1 < \text{len}(s)$

$\Rightarrow \Delta = \Delta \cup \{(s[0], s[1]) \in R1^+\}$

$|| \Delta = \Delta \cup \{(s[0], s[1]) \in R2^+\}$

$|| \Delta = \Delta \cup \{(s[0], s[1]) \in R3^+\}$

Δ_1

$1 < \text{len}(s)$
 $(s[0], s[1]) \in R1^+$

Δ_2

$1 < \text{len}(s)$
 $(s[0], s[1]) \in R2^+$

Δ_3

$1 < \text{len}(s)$
 $(s[0], s[1]) \in R3^+$

Example 1: exploring Δ_1 — Steps 4 & 5

Exploring Δ_1 only (Δ_2, Δ_3 omitted).

Step 4 — SplitCycleLen

Premise(s): $((R1, R2, R3), s, 2) \in C$

$\Rightarrow 2 < \text{len}(s)$ (case 1 — explored)

$\parallel 2 = \text{len}(s)$ (case 2 — omitted)

Δ_1'
 $2 < \text{len}(s)$
 $(s[0], s[1]) \in R1^+$

Step 5 — UnrollCycle

Premise(s): $((R1, R2, R3), s, 2) \in C$ and $2 < \text{len}(s)$

$\Rightarrow (s[1], s[2]) \in R1^+ \parallel R2^+ \parallel R3^+$

Δ_{11}

$2 < \text{len}(s)$
 $(s[0], s[1]) \in R1^+$
 $(s[1], s[2]) \in R1^+$

Δ_{12}

$2 < \text{len}(s)$
 $(s[0], s[1]) \in R1^+$
 $(s[1], s[2]) \in R2^+$

Δ_{13}

$2 < \text{len}(s)$
 $(s[0], s[1]) \in R1^+$
 $(s[1], s[2]) \in R3^+$

We explore Δ_{12} only (Δ_{11}, Δ_{13} omitted).

Example 1: exploring Δ_1 — Step 6

Step 6 — SplitCycleLen

Premise(s): $((R1, R2, R3), s, 3) \in C$

$\parallel$ $3 < \text{len}(s)$ (case 1 — omitted)

$\implies$ $3 = \text{len}(s)$ (case 2 — explored)

Δ_{12}

$2 < \text{len}(s)$

$(s[0], s[1]) \in R1^+$

$(s[1], s[2]) \in R2^+$

- **All cycle-related rules are now saturated** — assuming a guard on **SplitCycleLen** that we haven't reached the end of the cycle (no constraint $\text{len}(s) = n$ for a constant n in Δ).
- We can't apply **UnrollCycle**, whose premise $\text{cnt} < \text{len}(s)$ — here $3 < \text{len}(s)$ — is not in Δ , since we have $3 = \text{len}(s)$.
- Once we also saturate the original relational-solver rules, we can conclude **sat**.
- Congruence-closure reasoning concludes $s[0] = a, s[1] = b, s[2] = a$, which gives our model of R .

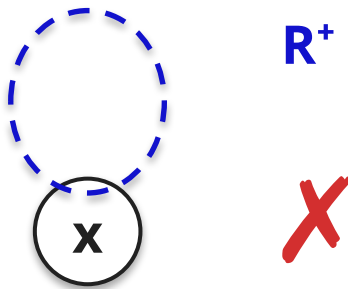
Model: $s[0] = a, s[1] = b, s[2] = a$

Example 2: unsat instance

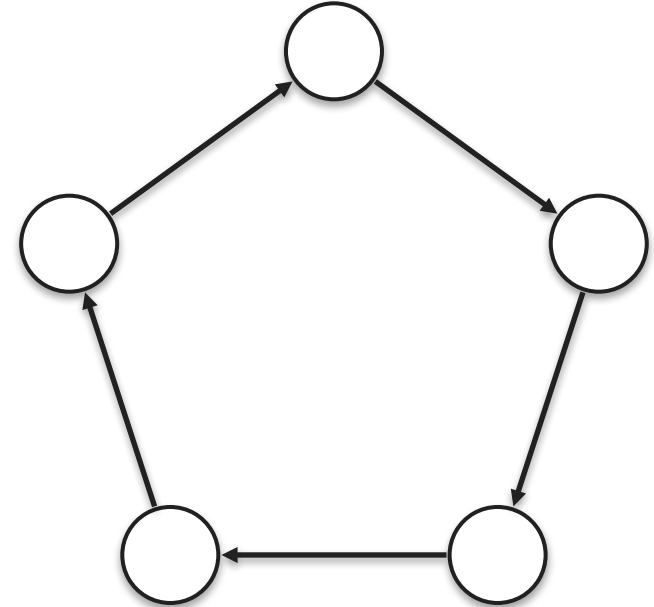
$\forall x : \text{Int. } \neg ((x, x) \in R^+)$

$\neg \text{acyclic}(R)$

i.e., R has a cycle, but R^+ has no self-loop



no self-loop (x, x) exists in R^+



a cycle in R

Example 2: derivation

Step 1 — InstCycle

Premise(s): $\neg \text{acyclic}(R)$

$\Rightarrow C = \{((R), s, 1)\}, \Delta = \{\text{len}(s) > 1, \text{minimal}((R), s)\}$

Step 2 — SplitCycleLen

Premise(s): $((R), s, 1) \in C$

$\Rightarrow 1 < \text{len}(s) \parallel 1 = \text{len}(s)$ (case 1)

Step 3 — UnrollCycle

Premise(s): $((R), s, 1) \in C$ and $1 < \text{len}(s)$

$\Rightarrow (s[0], s[1]) \in R^+$

Step 4 — SplitCycleLen

Premise(s): $((R), s, 2) \in C \Rightarrow$ explore both cases:

Case $2 < \text{len}(s)$:

$\Delta = \Delta \cup \{2 < \text{len}(s)\}$

continue: Steps 5–7 $\rightarrow$ (next slide)

$$\Delta_0 = \{\text{len}(s) > 1, \text{minimal}(s)\}$$

$$\Delta_0' = \{\text{len}(s) > 1, \text{minimal}(s)\}$$

$$\Delta_1 = \{1 < \text{len}(s), (s[0], s[1]) \in R^+\}$$

Case $2 = \text{len}(s)$:

$\Delta = \Delta \cup \{2 = \text{len}(s), s[0] = s[1]\}$

($s[0] = s[1]$ by definition of minimal)

$\Rightarrow (s[0], s[0]) \in R^+$ contradicts $\forall x. \neg((x, x) \in R^+)$

$\Rightarrow$ UNSAT — close this case

Example 2: case $2 < \text{len}(s)$

From Step 4 (case $2 < \text{len}(s)$): $\Delta \supseteq \{2 < \text{len}(s), (s[0], s[1]) \in R^+\}$

Step 5 — UnrollCycle

Premise(s): $((R), s, 2) \in C$ and $2 < \text{len}(s)$

$\implies \Delta = \Delta \cup \{(s[1], s[2]) \in R^+\}$

Step 6 — JoinUp

Premise(s): $(s[0], s[1]) \in R^+$ and $(s[1], s[2]) \in R^+$

$\implies \Delta = \Delta \cup \{(s[0], s[2]) \in R^+\}$

Step 7 — ContrMinimal

Premise(s): $((R), s, 3) \in C$, $0 \leq 0 < 2 < 3$, $(s[0], s[2]) \in R^+$

$\implies$ **UNSAT — close this case**

Both cases of Step 4 close $\implies$ Example 2 is unsat

Example 3: SC litmus test (§4.3.1)

Do the one-thread SC litmus-test templates forbid every MCM-violating execution? → expect unsat

po program order **rf** reads-from **fr** from-reads **mo** modification order

Litmus-test templates (irreflexive):

irreflexive(po⁺), irreflexive(rf), irreflexive(fr), irreflexive(mo)

irreflexive(po⁺ ⋈ rf), irreflexive(po⁺ ⋈ fr), irreflexive(po⁺ ⋈ mo)

Single thread:

is_singleton(Thread)

thread(x) = thread(y) $\iff$ x = y $\vee$ (x,y) $\in$ po⁺ $\vee$ (y,x) $\in$ po⁺

Identity: $\forall a. (a, a) \in \text{iden}$

Goal: $\neg \text{acyclic}(\text{po} \cup \text{rf} \cup \text{fr} \cup \text{mo})$

Example 3: derivation — setup & first unroll

Step 1 — Saturate (ElimTClos)

$\Rightarrow rf^+ \approx rf, fr^+ \approx fr$ (rf, fr relate disjoint sets)

Step 2 — skipped (no new constants yet)

Step 3 — InstCycle

Premise(s): $\neg \text{acyclic}(po \cup rf \cup fr \cup mo)$

$\Rightarrow C \cup = ((po, rf, fr, mo), s, 1); \Delta \cup = \{\text{len}(s) > 1, \text{minimal}((po, rf, fr, mo), s)\}$

Step 3 — SplitCycleLen (cnt = 1)

$1 \approx \text{len}(s) : \text{unsat} \text{ — conflicts } \text{len}(s) > 1$ (close)

$1 < \text{len}(s) : \text{continue}$

Step 3 — UnrollCycle (cnt = 1)

$\Rightarrow (s[1], s[2]) \in po^+ \parallel rf^+ \parallel fr^+ \parallel mo^+$ (4 cases; explore po^+ , others similar)

Example 3: case $(s[1], s[2]) \in po^+$

Step — SplitCycleLen (cnt = 2)

Premise(s): $((po, rf, fr, mo), s, 2) \in C \implies$ explore both cases:

Case $2 \approx \text{len}(s)$:

$s[1] = s[2]$ (by minimal)

$\implies (s[1], s[1]) \in po^+$

step 2: instantiate iden at $s[1]$

$\implies (s[1], s[1]) \in \text{iden}$

irreflexive(po^+): $po^+ \sqcap \text{iden} \approx \emptyset$

$\implies$ **UNSAT — close this case**

Case $2 < \text{len}(s)$:

continue $\implies$ UnrollCycle (cnt = 2)

$(s[2], s[3]) \in po^+ \parallel rf^+ \parallel fr^+ \parallel mo^+$

explore $(s[2], s[3]) \in rf^+$ (others similar)

$\rightarrow$ next slide

Example 3: case $(s[2], s[3]) \in rf^+$

Step 1 — Join Up

$\Rightarrow (s[1], s[3]) \in (po^+ \bowtie rf)$

Step 2 — instantiate thread at $s[1], s[3]$ (single thread)

$thread(s[1]) = thread(s[3]) \Rightarrow$

$s[1] = s[3] \vee (s[1], s[3]) \in po^+ \vee (s[3], s[1]) \in po^+ \text{ (split 3)}$

Δ

$rf^+ \approx rf$

$2 < len(s)$

$(s[1], s[2]) \in po^+$

$(s[2], s[3]) \in rf^+$

① $s[1] = s[3]$: with $(s[1], s[3]) \in (po^+ \bowtie rf) \Rightarrow$ contradicts $irreflexive(po^+ \bowtie rf)$. **UNSAT**

② $(s[1], s[3]) \in po^+$: ContrMinimal — non-consecutive elements related by a transitive closure $\Rightarrow$ smaller cycle.
UNSAT

③ $(s[3], s[1]) \in po^+$: TClos Up II $\Rightarrow (s[3], s[2]) \in po^+$; Join Up $\Rightarrow (s[3], s[3]) \in (po^+ \bowtie rf) \Rightarrow$ contradicts $irreflexive(po^+ \bowtie rf)$. **UNSAT**

All other cases similar $\Rightarrow$ Example 3 is unsat